PROBLEM SET 6

Revision Problem set with complex problems

1. A company has four share-holders A, B, C and D. A owns 20, B owns 30, C owns 40 and D owns 10 shares. A policy can be passed only if share-holders holding 60% or more of the total shares vote for it. Assuming that **A**, **B**, **C** and **D** denote binary variables indicating whether the corresponding share-holder has supported a particular policy or not, obtain a Boolean expression for the condition (**P** = 1) for a policy to get accepted.

2. The board of directors of a bank consists of two directors (D1, D2) and a managing director (MD). The vault of the bank can be opened on working days either by any two directors together or by the managing director. On Saturdays, any one of the directors and the managing director can together open the vault, while on Sundays, all three have to be present to open the vault. Assign binary variables **X**, **Y** and **Z** to indicate a working day, Saturday and Sunday respectively, and obtain a Boolean expression for the condition to be satisfied for opening the vault (**V**) in terms of the binary input variables **D1**, **D2**, **MD**, **X**, **Y** and **Z**.

3. A student about to graduate from Shiv Nadar University has to decide whether he should accept the job offer he has received through campus interviews, on the basis of

(a) Pay package (Excellent/Good/Poor),
(b) Location of the company (Metropolitan/Remote),
(c) Support for higher education given by the company (Full/Partial/Nil),
(d) His Friends also getting offer from the same company (Yes/No).

He will accept the offer either if the Pay package is Excellent and the company is located in a Metropolitan area, irrespective of other conditions, or if the Pay package is not Poor and the company is located in a Metropolitan area and at least Partial Support is given by the company for higher education. He will accept an Excellent Pay package from a company in a Remote location only if some of his friends have also been offered jobs there. He will accept even a Good Pay package from a company in a Remote location if the company gives Full Support for higher education and some of his friends have also been offered jobs there.

Assign binary variables **PE** (Excellent Pay), **PP** (Poor Pay) **LR** (Remote Location), **SF** (Full Support), **SP** (Partial Support) and **FG** (Friends Getting offer) to denote the items he has to consider, and express the output **A** (**A** = 1 indicating acceptance) as a Boolean function of these variables.

4. A student has to decide whether he will buy the textbook prescribed for a course he is going to take, based upon the following considerations:

(a) cost of the book – LOW/MEDIUM/HIGH,
(b) quality of the teacher – EXCELLENT/GOOD/AVERAGE,
(c) quality of lecture notes – GOOD/AVERAGE,

(d) the book being useful in other courses also – YES/NO,

He would like to buy the book if its cost is LOW. However, either if the teacher is EXCELLENT or if the teacher is reasonably GOOD and gives GOOD lecture notes, he feels that he may not have to buy the book. But if the teacher is AVERAGE, and the book is useful in other courses also, he would prefer to buy the book whatever its cost. If the book is useful in other courses, he would buy it in any case provided its cost is MEDIUM.

Assign binary variables **CL** (Low Cost), **CM** (Medium Cost) **QE** (Excellent Teacher), **QG** (Good Teacher), **LG** (Good Lecture notes) and **BU** (Book Useful) to denote the items he has to consider, and obtain a Boolean expression for the output **Y** representing the decision to buy the book.

5. A robot has ***four*** permitted directions of movement and ***three*** possible speed settings. Let the direction control and the speed control be applied through ***two bits*** each:

D1D0 = 00(FORWARD) / 11(REVERSE) / 01(RIGHT) / 10(LEFT), and
S1S0 = 00(OFF) / 01(LOW) / 10(MEDIUM) / 11(HIGH).

It is desired to have a protection system to ensure that the robot can move at HIGH speed for FORWARD movement only, and REVERSE only at LOW speed. This will have to be achieved by generating an output bit **P** which should go HIGH and shut off the power to the robot if any of these conditions is violated. Write down the Boolean expression for the output **P** in terms of **D1**, **D0**, **S1**, **S0**.

6. A combinational circuit has to provide automatic control of a car air-conditioner, consisting of a Compressor, a Heater and a 2-speed Fan, depending on whether Temperature (**T**) and Moisture (M) lie in the **LOW**, **MEDIUM** or **HIGH**. The different ranges of Temperature and Moisture will be indicated by the binary variables **TH**, **TL**, **MH** and **ML**, and the Control Outputs, by **C** (1 Compressor ON), **H** (1 Heater ON) and **F1F0** (00 Fan OFF, 01 Fan speed LOW, 10 Fan speed HIGH). Obtain Boolean expressions for the 4 Output variables **C**, **H**, **F1** and **F0** in terms of **TH**, **TL**, **MH** and **ML**, to implement the control logic given in the following table:

	LOW Moisture **MH = 0, ML** = 1	**MEDIUM Moisture** **MH = 0, ML** = 0	HIGH Moisure **MH = 1, ML** = 0
LOW Temperature **TH = 0, TL** = 1	Heater OFF Compressor OFF Fan OFF	Heater ON Compressor OFF Fan speed LOW	Heater ON Compressor OFF Fan speed HIGH
MEDIUM Temperature **TH = 0, TL** = 0	Heater OFF Compressor ON Fan OFF	Heater OFF Compressor OFF Fan OFF	Heater ON Compressor OFF Fan speed LOW
HIGH Temperature **TH = 1, TL** = 0	Heater OFF Compressor ON Fan speed LOW	Heater OFF Compressor ON Fan speed HIGH	Heater OFF Compressor ON Fan speed HIGH

7. The control unit of a chemical process is required to control the temperature and the pressure inside a reactor by ON/OFF control of a Heater (**H**) and an inlet Valve (**V**), depending on the Temperature (LOW/NORMAL/HIGH) and Pressure (LOW/NORMAL/HIGH) according to following logic:

Heater is ON (**H** = 1) if the Temperature is LOW and the Pressure is not HIGH.

Valve is Open (**V** = 1) if Pressure is LOW and Temperature is not LOW.

In addition, the control unit has also to sound an Alarm (A) if the temperature and the pressure are either both LOW or both HIGH. Assign two binary variables **TH** and **TL** to represent Temperature ranges – LOW (**TH** = 0, **TL** = 1), NORMAL (**TH = 0, TL** = 0) and HIGH (**TH = 1, TL** = 0), and two binary variables **PH** and **PL** to represent Pressure ranges – LOW (**PH = 0, PL** = 1), NORMAL (**PH = 0, PL** = 0) and HIGH (**PH = 1, PL** = 0). Obtain Boolean expressions for the output variables **H**, **V** and **A** in terms of **TH**, **TL**, **PH** and **PL**.

8. An intelligence agency uses a special code for storing numerical data for the sake of data security, representing each 4-bit number (N) by a 4-bit binary code PQRS according to the following scheme:

For $4 \geq N \geq 0$, PQRS = 13 – N, e.g. if N = 2, PQRS = 1011, for $11 \geq N \geq 5$, PQRS = N – 3, e.g. if N = 6, PQRS = 0011,
for $13 \geq N \geq 12$, PQRS = N + 2, e.g. if N = 12, PQRS = 1110, and for $15 \geq N \geq 14$, PQRS = 15 – N, e.g. if N = 15, PQRS = 0000.

A DECODER has to be designed for obtaining the normal binary representation $B_3B_2B_1B_0$ of N from its PQRS representation.

Construct the K-maps for the 4 bits B_3, B_2, B_1, B_0 in terms of P, Q, R, S.

Obtain the Boolean expressions for B_3, B_2, B_1, B_0 in the sum of products form.

9. A combinational circuit has to be designed for performing the operation z = (x + 1)y + 2, where both x and y are positive integers in the range 0–3 and are represented by 2-bit binary codes X_1X_0 and Y_1Y_0 respectively. z is represented by the four-bit binary code $Z_3Z_2Z_1Z_0$.

Construct Karnaugh maps for the 4 output bits Z_3, Z_2, Z_1, Z_0.

Obtain the minimal Boolean expressions for Z_3, Z_2, Z_1, Z_0 in the sum of products form.